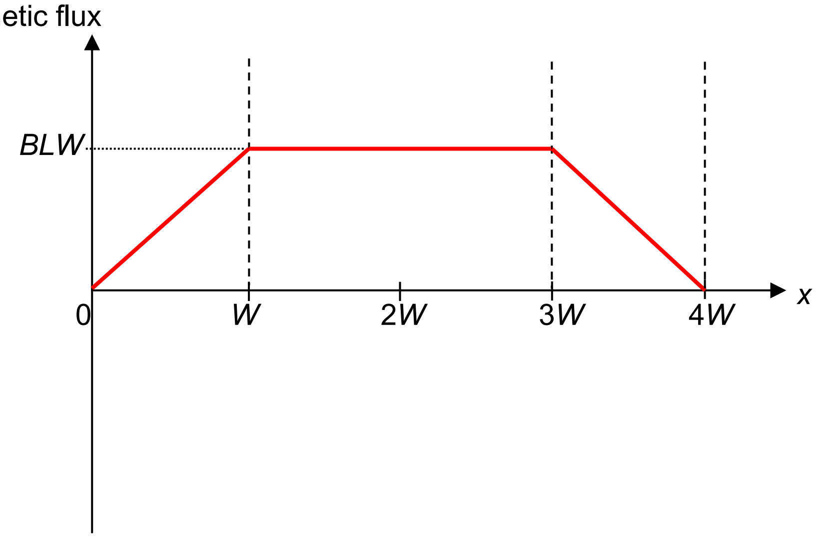
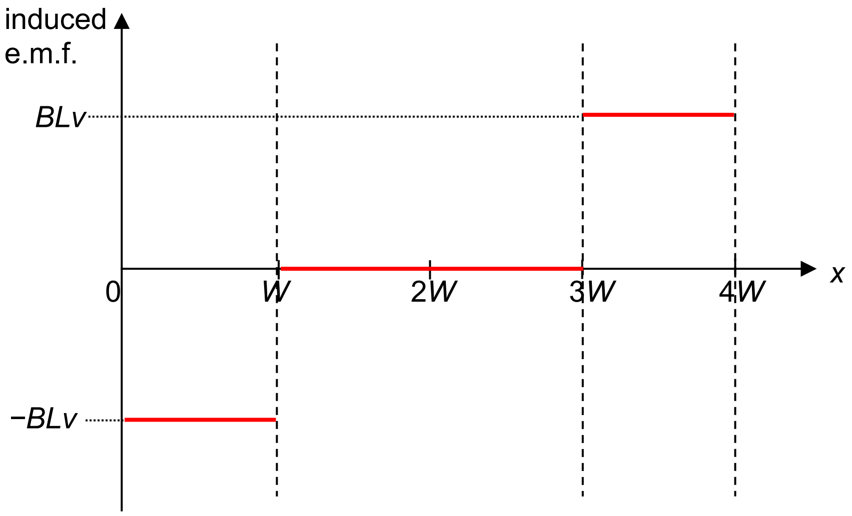
8 (a) (i)



[B1] correct shape

[B1] correct expression for maximum magnetic flux linkage

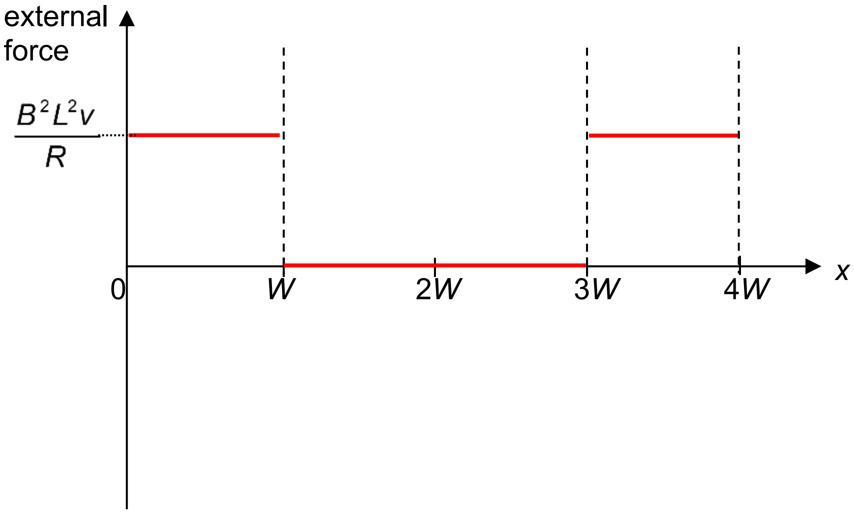
(ii)



[B1] correct shape

[B1] correct expression for maximum induced e.m.f.

(iii)



[B1] correct shape

[B1] correct expression for maximum external force applied

$$|F_{ext}| = |F_B| = BIL = B\left(\frac{\varepsilon}{R}\right)L = B\left(\frac{BLv}{R}\right)L = \frac{B^2L^2v}{R}$$

F_B is directed to the left as the frame is entering and exiting the field, hence to maintain constant speed, F_{ext} is directed to the right, hence positive value.

(b) **(i)** Faraday's Law of Electromagnetic Induction states that the induced e.m.f is directly proportional to the rate of change of magnetic flux linkage. [B2]

(ii) (Any 2)

Area enclosed by the coil

magnetic flux density

Speed of rotation of the coil / angular velocity of the coil

[B1] for any one correct

(iii) When the coil rotates at constant angular velocity, the magnetic flux density perpendicular to the coil varies sinusoidally. [B1] (alt: Area perpendicular to the flux)

As magnetic flux density is directly proportional to magnetic flux (linkage), it also varies sinusoidally. [B1]

Since induced e.m.f. is directly proportional to rate of change of magnetic flux (linkage), (Or by Faraday's Law) it also changes sinusoidally. [B1]

(c) **(i)** $F = NBIL\sin\theta$

$$B = \frac{F}{NIL\sin\theta}$$

$$= \frac{0.35}{50(7.0)(9.0\times10^{-2})(\sin 62°)} \qquad [M1]$$

$$= 0.0126 \text{ T} \qquad [A1]$$

If forget to include N turns, but have sin 62, give 1 mark

(ii) (From Fleming's left-hand rule) Force on PQ is upwards, hence the reading decreases. [B1]

Can BOD "out of paper"

(iii) Coil will rotate (or oscillate) about the vertical axis due to the couple provided the magnetic forces acting on the vertical sides. [B1]

(d) **(i)** To reduce energy losses or eddy currents [B1]

or To improve efficiency

(ii)

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\frac{V_s}{102} = \frac{600}{30}$$

$$V_s = 2040 \text{ V} \qquad [B1]$$

(iii)

$$I_{rms} = \frac{V_{rms}}{R}$$

$$= \frac{2040/\sqrt{2}}{98}$$

$$= 14.7 \text{ A} \qquad [B1]$$